Reward-Channel Separability in Multimodal Chart RLVR: A Cross-Rollout Bonus and a Negative-Sample Caveat

Anonymous ACL submission

Abstract

Chart question answering requires a model to parse the visual structure, extract the relevant data, and reason to a final answer. Whether these components respond independently to different training signals, however, is not well understood. We study this with two opposing interventions: Hierarchical Correct-Path Consistency (**HCPC**), a cross-rollout bonus rewarding consistent extraction and diverse reasoning among correct responses, and Negative Sample Reinforcement (**NSR**; **?**), which removes gradient signal from correct responses entirely. We evaluate on ChartQA (in-distribution) and the held-out ChartFC benchmark, which is excluded from training and serves as our out-of-distribution (OOD) test. NSR collapses format compliance from 96.8% to 31.6% on ChartQA with no gain in answer accuracy over GRPO. Per-tag analysis traces the failure to the structural tags reward training had to teach, pointing to reward-signal starvation as the cause. HCPC achieves the best Pass@4 on ChartQA (0.828 *vs.* 0.816 for GRPO) but degrades on ChartFC (0.918 *vs.* 0.922 for GRPO), where reliance on structured extraction becomes a liability. Combining the two, NSR+HCPC recovers GRPO-level accuracy on ChartFC while format compliance barely recovers. The fact that accuracy returns but format does not suggests the two are distinct behavioral channels, and that reward update rules from single-component settings should be re-evaluated before applying them to richer reward landscapes. We will make our code and data public.

1 Introduction

Recent chart-reasoning vision-language models are trained via reinforcement learning with verifiable rewards (RLVR), typically using GRPO with multi-component rewards. Answering questions about charts requires two sequential operations that are rarely treated as distinct (**????**). A model must first extract the underlying data table from the visual layout, then reason over that table to reach an answer. The first operation has one correct outcome; the second admits many valid paths. Current training pipelines collapse this distinction. The dominant recipe applies GRPO (**?**) with verifiable rewards across chart-type identification, table reconstruction, process conformity, and answer accuracy (**????**), reinforcing all correct rollouts uniformly regardless of which operation they got right or how. The practical cost is gradual: the policy over-commits to whichever solution paths dominate the training distribution, eroding the diversity that supports robustness under distribution shift and coverage at higher K in Pass@K (**?**).

We propose Hierarchical Correct-Path Consistency (**HCPC**), a cross-rollout bonus that treats the two operations differently. Among rollouts that get both the table and the answer right, HCPC rewards consistent table extraction and diverse reasoning traces. Consistency is appropriate at the extraction layer, where there is one correct table and reliable recovery is the goal. Diversity is appropriate at the reasoning layer, where multiple paths to the same answer reflect genuine competence. A ground-truth-anchored filter ensures the bonus targets correct-path quality rather than surface agreement among arbitrary rollouts.

To isolate what HCPC is and is not doing, we introduce a contrastive diagnostic: Negative Sample Reinforcement (NSR; **?**), which masks gradient on correct rollouts and penalizes only incorrect ones. NSR was motivated by mitigating reward hacking and overconfident convergence, though its mechanism may also encourage exploration and diversity. In single-component math RLVR it works (**?**). In chart RLVR it does not: porting it unchanged collapses format compliance, specifically the structured XML tags and table markup the pipeline depends on, on both ID and OOD data, with no gain in answer accuracy. Per-tag analysis traces the

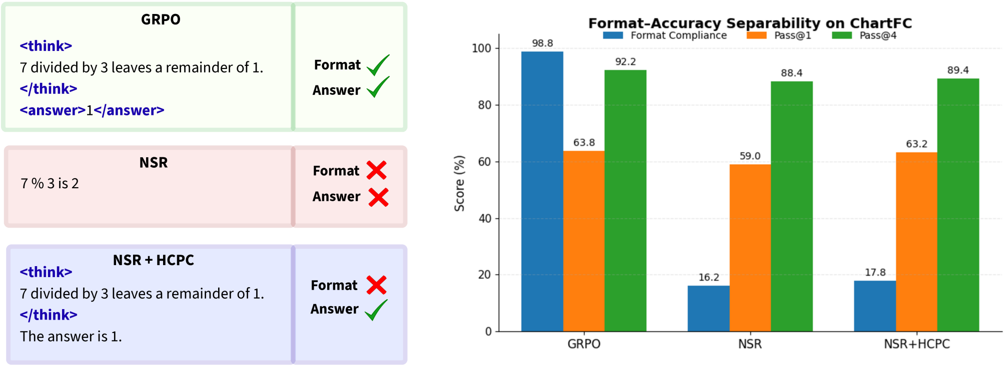

Figure 1: Format and answer accuracy are separable channels in chart RLVR. **Left:** a stylized rollout under each training regime. GRPO emits the required structural tags and the correct answer; NSR drops both; NSR+HCPC recovers the answer while the structured tags remain collapsed. **Right:** ChartFC (OOD) metrics for the same three regimes. Adding HCPC on top of NSR restores Pass@1 and Pass@4 to near GRPO levels, but format compliance barely moves from NSR's collapsed value (16.2% to 17.8%). Accuracy returns; format does not. The dissociation is direct evidence that these two behaviors respond independently to training interventions.

failure to exactly the structural tags reward training had to teach, pointing to reward-signal starvation. The failure is informative precisely because it is clean: something about the multimodal, multi-component reward landscape breaks an assumption that held in the simpler setting. Adding HCPC to NSR recovers competitive answer accuracy out-of-distribution while format compliance stays near zero. Accuracy returns; format does not. This dissociation is the central finding of the paper: format compliance and answer accuracy are distinct behavioral channels in chart RLVR, and they can be manipulated independently. The implication is that reward channels in this setting are separable and should be designed with that separability in mind. Our contributions are as follows:

1. **HCPC (§3.2, §5.2).** A cross-rollout bonus applying level-specific pressure to table extraction and reasoning separately. HCPC improves rollout-set robustness (Pass@4, all-rollouts-correct rate, format compliance) while maintaining parity with the GRPO baseline on single-sample accuracy.

2. **NSR transferability analysis (§5.3).** Porting NSR unchanged into our multi-component chart-reasoning training pipeline collapses format compliance without accuracy gain. Per-tag analysis identifies reward-signal starvation as the mechanism, localized to structural tags the base model does not emit unprompted.

3. **Separability of format and accuracy (§5.4).** Adding HCPC to NSR recovers out-of-distribution accuracy all the way back to the GRPO baseline, while format compliance barely moves from NSR's collapsed level. One channel snaps back; the other does not. This is direct evidence that the two are distinct behavioral channels that respond independently to training interventions. We argue reward components in chart RLVR should be designed with this separability explicit.

2 Related Work

Chart reasoning with RLVR. Most current chart-reasoning VLMs are trained with GRPO and multi-component verifiable rewards. Chart-RVR (?) set the template, combining rewards for format, chart-type, table reconstruction, process conformity, and answer accuracy. Recent work extends this along orthogonal axes: reasoning-trace supervision and CoT distillation (??), richer training data (??), grounding reflection (?), and agentic pipelines that wrap base models rather than modifying training (???). Our work sits next to these. We utilize Chart-RVR's reward backbone and ask a different question: how the choice of advantage rule and a cross-rollout reward interact with it.

Advantage-rule variants in RLVR. GRPO (?) treats correct and incorrect rollouts symmetrically, normalizing rewards within a group. **?** split this into positive- and negative-sample reinforcement (PSR and NSR), and show that NSR alone, masking gradients on correct rollouts and penalizing only incorrect ones, matches or beats GRPO on math benchmarks. Related work studies RLVR in adjacent settings (????). All assume a single scalar reward, the setting NSR was designed for. We are the first to port NSR into a multi-component, multimodal reward landscape, and to identify a clean failure mode: structural rewards saturate early, pushing well-formed but answer-wrong rollouts above NSR's correctness threshold, which masks them and starves the structural side of positive gradient.

Group-level signals and rollout-set diversity. Self-consistency (?) is the closest existing cross-rollout signal, but operates at inference, aggregating K samples by majority vote for calibration, not training. ETTRL (?) preserves diversity at test time via entropy control. HCPC differs in three ways: it acts at training time as a reward bonus; it is anchored to ground-truth correctness, not majority agreement, so consistently wrong groups get no bonus; and it applies different objectives at different levels, consistency for table extraction and diversity for reasoning. We use Pass@K (?) as our coverage diagnostic; HCPC's level-specific design targets the higher-K regime where symmetric advantage rules fall behind.

Format and accuracy as separable behaviors. The dissociation we find, where format compliance and answer accuracy move in opposite directions under the same intervention, has no direct precedent in chart RLVR we are aware of. The closest adjacent observation is that CoT prompting can degrade VLM accuracy in chart settings (?), suggesting structured reasoning and correctness already trade off through some channel. Our result is stronger than a tradeoff: NSR+HCPC recovers GRPO-level accuracy while format compliance stays at NSR's collapsed level. This makes reward-component independence a real design choice for VLM RLVR, not a side note.

3 Methodology

We build on Chart-RVR (?) as a reward backbone and introduce two interventions: HCPC (Hierarchical Correct-Path Consistency), a group-level bonus rewarding level-specific structure among correct rollouts, and NSR (?), a contrastive gradient-masking strategy that we port to chart RLVR as a diagnostic.

3.1 Preliminaries: Chart-RVR Reward Backbone

We adopt Chart-RVR's per-rollout reward structure, which supervises each output component independently. Rollouts are expected in the format <think><type/> <table/> reasoning</think> <answer/>. Four components are active: R_{fmt} (structural compliance with the required XML template), R_{acc} (answer correctness against y^*), R_{tbl} (table extraction quality assessed by partial structural alignment against T^*), and R_{type} (exact match of the predicted chart type against the ground-truth type c^*). Their sum gives the base reward:

$$R_{\text{base}}(o_i) = R_{\text{fmt}} + R_{\text{acc}} + R_{\text{tbl}} + R_{\text{type}}.$$

For answer evaluation, we intentionally decouple extraction of $\hat{y}_i$ from full-format validity. We read $\hat{y}_i$ from the first <answer>...</answer> span if one is present, even if other parts of the output violate the XML schema; if the answer span is missing or malformed, we set $\hat{y}_i{=}\varnothing$ and mark the rollout incorrect. Thus a rollout can be answer-correct but format-incorrect, which is precisely the operational basis of the format/accuracy dissociation analyzed later. By contrast, HCPC's correct-path filter in §3.2.1 uses the stricter requirement that the entire response be structurally parseable.

Chart-RVR's other reward components (length, token-count, and process-conformity) are disabled: the first two are noisy on this backbone because the base model's reasoning yields variable lengths and duplicate tags, and process-conformity (similarity to the GT reasoning chain) conflicts with our diversity objective.

3.2 HCPC: Hierarchical Correct-Path Consistency

HCPC augments R_{base} with a group-level bonus computed exclusively over correct rollouts. The design enforces *level-specific coherence*: consistency where one factually correct answer exists (table extraction) and diversity where multiple paths

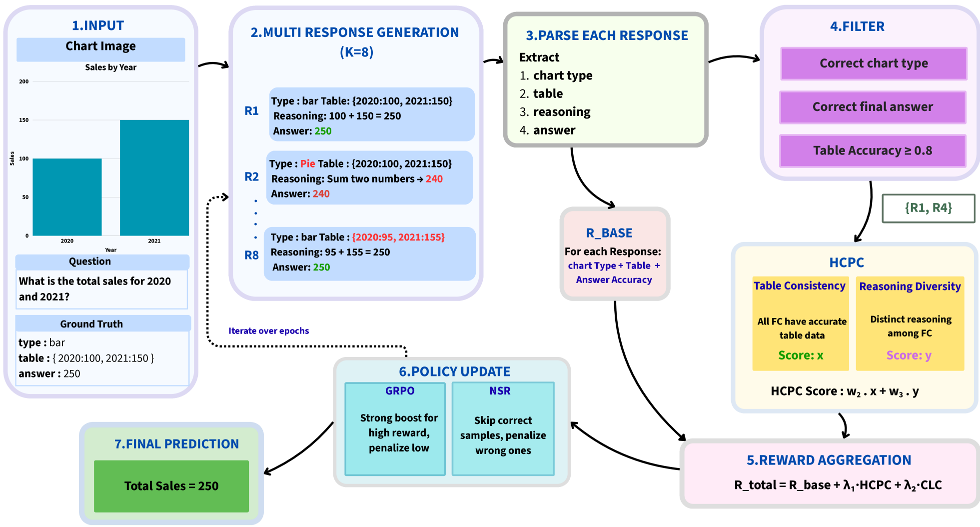

Figure 2: Training pipeline. The multi-component reward supervises each rollout independently through format, accuracy, table, and chart-type rewards. HCPC adds a group-level bonus computed over rollouts that pass the correct-path filter (G^+), rewarding consistent table extraction together with diverse reasoning. We compare two advantage estimators over the same per-rollout reward: standard GRPO and NSR, which masks the gradient of correct rollouts.

are valid (reasoning steps). Cross-rollout statistics are conditioned on ground-truth correctness rather than majority agreement (?), preventing shared extraction errors from producing a spuriously high consistency signal.

3.2.1 Correct-Path Filtering

Let $G = \{o_1, \dots, o_K\}$ be the rollouts for a prompt with ground truth (T^*, y^*). We define the *correct-path subset* $G^+ \subseteq G$ via three sequential gates.

Gate 1 (format). The rollout must be structurally parseable with all required tags and a valid table block.

Gate 2 (table fidelity). The extracted table $\hat{T}_i$ must satisfy $\mathrm{sim}(\hat{T}_i, T^*) \geq \delta$ with δ=0.8, using cosine similarity over MiniLM-L6 embeddings (?) on serialized table strings. This excludes rollouts that reach the correct answer from a misread table.

Gate 3 (answer). The predicted answer $\hat{y}_i$ must match y^*: relative error $\leq$5% for numeric answers, normalized substring match for string answers.

All three gates must pass. An answer-correct rollout with a wrong table has not followed a genuine reasoning path; rewarding its cross-rollout properties would propagate extraction errors into the group signal.

3.2.2 Level-Specific Group Metrics

On G^+ we compute three metrics targeting distinct trajectory levels. C_{type} is the fraction of rollouts in G^+ that predict the modal chart type $\hat{c}_{\text{mode}}$. For table and reasoning levels, let $\langle f \rangle_{G^+}$ denote the mean of $f(i,j)$ over all distinct pairs $i < j$ in G^+:

$$C_{\text{table}} = \big\langle \mathrm{sim}(\hat{T}_i, \hat{T}_j) \big\rangle_{G^+},$$

$$D_{\text{reason}} = 1 - \big\langle \mathrm{sim}(\hat{w}_i, \hat{w}_j) \big\rangle_{G^+}.$$

C_{table} measures pairwise agreement among extracted tables; correct rollouts should converge on the one ground-truth reading. D_{reason} is pairwise dissimilarity among reasoning strings $\hat{w}_i$; high values indicate diverse inferential paths to the same answer. C_{type} and C_{table} enforce convergence where the correct answer is unique; D_{reason} enforces divergence where multiple paths are valid. All similarities use cosine similarity over MiniLM-L6 embeddings.

3.2.3 Reward Integration

The consistency component of the group bonus is shared equally among all rollouts in G^+:

$$B_{\text{cons}} = \frac{|G^+|}{K}\big(w_{\text{type}}\, C_{\text{type}} + w_{\text{table}}\, C_{\text{table}}\big).$$

The $|G^+|/K$ prefactor scales the consistency bonus by the correct-path fraction, so the signal grows with the share of rollouts the model already gets right and vanishes when the group is mostly incorrect. The diversity component is assigned per-rollout via a uniqueness score $u_i = 1 - \bar{s}_i$ for $i \in G^+$, where

$$\bar{s}_i = \frac{1}{|G^+| - 1} \sum_{j \in G^+,\, j \neq i} \mathrm{sim}(\hat{w}_i, \hat{w}_j)$$

is rollout i's mean cosine similarity to the other reasonings in G^+. The rollout average of u_i over G^+ equals D_{reason}, so u_i is the per-rollout decomposition of the group-level diversity. For groups with $|G^+| \geq 2$, the per-rollout HCPC reward is:

$$R_{\text{HCPC}}(o_i) = \begin{cases} B_{\text{cons}} + w_{\text{reason}}\, u_i & \text{if } o_i \in G^+, \\ 0 & \text{otherwise,} \end{cases}$$

with $w_{\text{type}}{=}1.0$, $w_{\text{table}}{=}2.0$, and $w_{\text{reason}}{=}1.5$. The total reward is:

$$R_i = R_{\text{base}}(o_i) + R_{\text{HCPC}}(o_i).$$

Choice of weights. Weights are fixed (not tuned) and scaled to the empirical range of each term. C_{table} gets the largest weight because it ranges in $[0, 1]$ and is the term most directly tied to correctness. C_{type} shares the same range but is near-ceiling on firing groups (chart-type saturates early), so a smaller weight avoids a near-constant offset. The per-rollout uniqueness u_i has a much smaller empirical range ($\bar{u}{\approx}0.05$ at $K{=}4$), so w_{reason} is raised above w_{type} to keep the diversity term non-trivial; even so, $w_{\text{table}} C_{\text{table}}$ dominates. A full grid sweep over $(w_{\text{type}}, w_{\text{table}}, w_{\text{reason}})$ is beyond our compute budget; we leave systematic ablation to future work. HCPC is silent when $|G^+|{<}2$; on ChartQA this occurs 28.8% of the time ($|G^+|{=}0$ on 17.2% of groups, $|G^+|{\geq}3$ on 56.6%).

3.3 Diagnostic Baseline: NSR

Negative Sample Reinforcement (**?**) pursues the same diversity goal as HCPC by the opposite route: zeroing gradient on correct rollouts and penalizing only incorrect ones. It improves Pass@K in single-component math RLVR (**?**); we port it unchanged to test transfer to multi-component reward.

Masking rule. After GRPO normalizes advantages within a group, we zero the advantage of any rollout with $R_i \geq \theta = 0.5\, R_{\max}$, where $R_{\max}$ sums all defined Chart-RVR ceilings whether or not each component is active: 11.0 base, 15.5 with HCPC ($11.0 + w_{\text{type}} + w_{\text{table}} + w_{\text{reason}}$), so $\theta{=}5.5$ and 7.75 respectively. With three base components inactive (§3.1), the threshold is conservative and masks only the highest-scoring rollouts. It is fixed, not group-relative.

NSR+HCPC. The HCPC bonus inflates R_i for rollouts in G^+, making them more likely to be masked; rollouts outside G^+ keep their negative gradient. HCPC membership thus indirectly selects which rollouts drive the policy update under NSR.

4 Experimental Setup

4.1 Data and Training Setup

Training uses 1,000 ChartQA (**?**) samples drawn from Chart-RVR's CoT training mixture (**?**). ChartQA is our in-distribution benchmark and ChartFC (**?**) is our held-out OOD benchmark; no ChartFC samples are used in training.

We fine-tune Qwen2.5-VL-3B-Instruct with LoRA ($r{=}8$, $\alpha{=}16$ on W_q, W_v) using AdamW at learning rate 1×10^{-6}, effective batch size 4 (per-device 1, gradient accumulation 4), $K{=}4$ rollouts per prompt at temperature 0.8 and top-$p{=}1.0$, over 2 epochs. No KL penalty is applied ($\beta_{\text{KL}}{=}0$). In our setup, the verifiable rewards themselves act as the main constraint against degenerate outputs, and LoRA (restricted to W_q, W_v with $r{=}8$) sharply limits how far the policy can drift from the base model, so an additional reference-policy KL term was empirically unnecessary on our 1,000-sample, 2-epoch budget. We evaluate five configurations: BASE (untuned Qwen2.5-VL-3B-Instruct), GRPO, GRPO+HCPC, NSR, and NSR+HCPC. GRPO uses the standard group-normalized advantage $\hat{A}_i = (R_i - \bar{R}) / \max(1, s_R)$; NSR additionally zeros out advantages on rollouts with $R_i \geq \theta$ (§3.3).

4.2 Evaluation Protocol

Evaluation samples 4 generations per test prompt at temperature 0.8 and top-$p{=}0.95$, the standard inference-time setting; training uses top-$p{=}1.0$ to keep the rollout distribution unrestricted during exploration. We evaluate on two disjoint 500-sample slices, one per benchmark, drawn from the official test split of each. **ChartQA** (**?**) provides the in-distribution slice; **ChartFC** (**?**) provides the held-out OOD slice, differing from ChartQA on task format (fact-checking vs. open-ended QA) and visual style. The two slices are reported separately and never mixed. We report Pass@1, unbiased Pass@4 (?), and format compliance Fmt%. Two additional metrics are central to this paper.

Per-tag emission rate. For each required tag (⟨think⟩, ⟨answer⟩, ⟨type⟩, ⟨table⟩, plus proper order), the fraction of rollouts emitting it. This localizes format collapse to specific structural components rather than reporting it in aggregate.

Table-dependence Δ_{tbl}.

$$\Delta_{\text{tbl}} = P(\hat{y} = y^* \mid \text{tbl parsed}) - P(\hat{y} = y^* \mid \text{no tbl}),$$

computed using *within-sample averaging*: for each test sample s we form the per-sample conditional means $\bar{p}_s^+ = \text{mean}_i\{\hat{y}_i = y^* : \text{tbl}_i \text{ parsed}\}$ and $\bar{p}_s^- = \text{mean}_i\{\hat{y}_i = y^* : \text{tbl}_i \text{ not parsed}\}$ over the K=4 rollouts of that sample. The slice-level Δ_{tbl} then averages each per-sample mean across the samples that have at least one rollout in the relevant condition. This keeps each test sample as the unit of observation (so paired-bootstrap CIs remain valid) while using all K rollouts to reduce per-sample noise, rather than relying on a single rollout. A higher Δ_{tbl} indicates the model's answer correctness depends more strongly on emitting a parseable table. The two arms of the conditional can move for different reasons; we unpack this in §5.2.

5 Results and Analysis

5.1 Main Results

Table 1 reports the five configurations on ChartQA and ChartFC. GRPO training substantially improves both answer accuracy and format compliance over the untuned Base model: Pass@1 rises from 0.518 to 0.630 on ChartQA and Fmt% from 29.8 to 96.8. GRPO+HCPC matches GRPO on Pass@1 (0.622 vs. 0.630) and pulls ahead at higher K (Pass@4 0.828 vs. 0.816), consistent with HCPC's design goal of improving rollout-set coverage rather than single-shot accuracy. GRPO retains the strongest table-dependence on both benchmarks (Δ_{tbl}=+0.265 on ChartQA, +0.160 on ChartFC), indicating that the un-augmented advantage rule already routes answer correctness through the extracted table; HCPC trades a small amount of this dependence for the rollout-set robustness gain. NSR collapses format compliance to 31.6% on ChartQA and 16.2% on ChartFC with no accuracy gain; the per-tag analysis in §5.3 traces this to reward-signal starvation on structural tags. NSR+HCPC recovers OOD accuracy to 0.632 (statistically tied with GRPO at 0.638; §5.4) while format compliance stays at 17.8%, the format/accuracy dissociation that is the central finding of this paper.

5.2 GRPO+HCPC: Rollout Robustness Without Accuracy Regression

HCPC is designed to strengthen correct-path quality by rewarding consistent extraction and diverse reasoning. The metrics that should reflect this are rollout-set robustness and Pass@K at higher K, not single-shot Pass@1.

Headline Pass@1 on ChartQA is statistically tied: GRPO 0.630 vs. GRPO+HCPC 0.622 (paired bootstrap −0.8 pp, CI [−3.8, +5.4], p=0.77, n_{boot}=10,000). On rollout-set robustness metrics, the direction favors HCPC: the fraction of test samples where all 4 rollouts are correct is 40.8% for HCPC vs. 38.6% for GRPO; the average correct rate per group is 0.628 vs. 0.620; Pass@4 is 0.828 vs. 0.816; format compliance is 98.2% vs. 96.8%. Table-dependence is the one direction where GRPO leads (Δ_{tbl}=+0.265 vs. HCPC's +0.144), which we unpack below. No individual difference reaches significance at n=500 (Pass@4 boot-p=0.48), but the joint direction across robustness metrics is consistent with the design hypothesis that a cross-rollout bonus preserves correct-path coverage without trading off single-shot accuracy. Figure 3 shows the Pass@K curves: HCPC matches GRPO at K=1 and pulls ahead at $K \geq 3$, the regime where rollout-set coverage matters.

Unpacking Δ_{tbl}: $P(\hat{y} = y^* \mid \text{tbl})$ is 0.625 for GRPO and 0.634 for HCPC (tied; also tied on the common-table intersection, §G). The gap comes entirely from the no-table arm: $P(\hat{y} = y^* \mid \text{no tbl})$ is 0.360 for GRPO and 0.490 for HCPC. The correct-path bonus makes the policy more robust to extraction failures rather than more dependent on the extracted table.

GRPO+HCPC OOD behavior. On ChartFC, GRPO+HCPC trails GRPO on most metrics (Pass@1 0.606 vs. 0.638, Pass@4 0.918 vs. 0.922, format 66.4% vs. 98.8%). Format is the largest drop: HCPC's extraction is less reliable on unfamiliar visual styles, and 34% of rollouts no longer emit a parseable table. Despite this, no-table accuracy holds up ($P(\hat{y} = y^* \mid \text{no tbl})$=0.605 for HCPC vs. 0.500 for GRPO on its few no-table

	ChartQA (in-distribution, n=500)				ChartFC (OOD)			
Method	Pass@1	Pass@4	Δ_{tbl}	Fmt%	Pass@1	Pass@4	Δ_{tbl}	Fmt%
Base	0.518	0.774	+0.005	29.8	–	–	–	–
GRPO	**0.630**	0.816	**+0.265**	96.8	**0.638**	**0.922**	**+0.160**	**98.8**
GRPO+HCPC	0.622	**0.828**	+0.144	**98.2**	0.606	0.918	+0.020	66.4
NSR	0.592	0.808	+0.041	31.6	0.590	0.884	−0.064	16.2
NSR+HCPC	0.574	0.800	+0.118	43.0	0.632	0.894	−0.003	17.8

Table 1: Main results. Δ_{tbl} is the conditional gain in answer correctness from emitting a parseable table; higher means stronger table-dependence. Bold marks column-best per benchmark. Pass@4 uses the unbiased estimator over 4 generations per prompt.

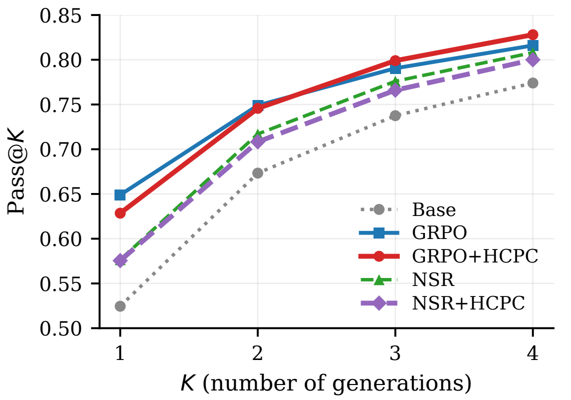

Figure 3: Pass@K on ChartQA (n=500). HCPC's gain over GRPO grows with K: tied at K=1, leading at K=3 and K=4. The direction is consistent with the hypothesis that a cross-rollout bonus preserves correct-path coverage at higher K. Individual differences are not significant at n=500 (Pass@4 boot-p=0.48).

Method	think	answer	type	table	order
Base	52.2	91.2	96.6	77.2	81.6
GRPO	97.4	97.6	100	98.6	97.4
GRPO+HCPC	98.8	98.2	100	99.6	98.6
NSR	57.4	94.8	96.6	72.2	85.4
NSR+HCPC	57.8	91.6	94.6	83.2	84.0

Table 2: Per-tag emission rate (%) on ChartQA. Column headers refer to the `<think>`, `<answer>`, `<type>`, `<table>` tags and the "proper order" check. NSR's collapse is concentrated in the structural tags (think, table) that the base model does not emit unprompted; HCPC under NSR partially recovers table but not think.

samples), so the Pass@1/Pass@4 gaps stay small (−3.2 and −0.4 pp). Δ_{tbl} drops to +0.020: the no-table arm is now large and still mostly correct, so the parseable-table condition no longer carries a sizable correctness premium. This is a genuine limitation of standalone GRPO+HCPC, complementary to the NSR-mitigation result in §5.4; fixes (chart-style augmentation) are left to future work.

5.3 NSR Breaks VLM Format

Switching the advantage estimator from GRPO to NSR, with no other changes, drops format compliance from 96.8% to 31.6% on ChartQA and from 98.8% to 16.2% on ChartFC. On ChartFC, GRPO leads NSR on Pass@4 by +3.8 pp (CI [+0.6, +7.2], p=0.028, paired bootstrap; McNemar exact p=0.034).

Table 2 localizes the collapse. NSR keeps the tags the base model already emits (⟨type⟩, ⟨answer⟩, proper order) and loses the ones GRPO has to teach (⟨think⟩, ⟨table⟩). NSR's ⟨think⟩ rate (57.4%) is essentially the base model's (52.2%) plus noise; ⟨table⟩ rate (72.2%) falls below the base model's (77.2%).

The mechanism is mechanical. R_{fmt}, the active structural reward, saturates quickly: emitting the required tags is a learned-once behavior that yields near-maximal contribution. With R_{fmt} saturated, the per-rollout R_i for a well-formed rollout reaches $\theta = 0.5\,R_{\max}$ regardless of answer correctness. NSR therefore masks *structure-good, answer-wrong* rollouts, leaving only *structure-bad, answer-wrong* ones to drive the policy. This removes the dominant positive gradient on structural-tag emission, and the policy drifts back toward the base model's unstructured output. We expect this failure mode to be absent in ?'s math experiments because text-math rewards are dominated by a single answer bit; the implicit format is either trivial or emerges from the answer-correctness signal itself.

5.4 Format and Accuracy as Separable Channels

On ChartFC, HCPC added to NSR improves every metric: Pass@1 0.590 → 0.632 (+4.2 pp), Pass@4 0.884 → 0.894 (+1.0 pp), Δ_{tbl} −0.064 → −0.003 (+0.061), and format 16.2% → 17.8%. The Δ_{tbl} shift from clearly negative to near zero indicates

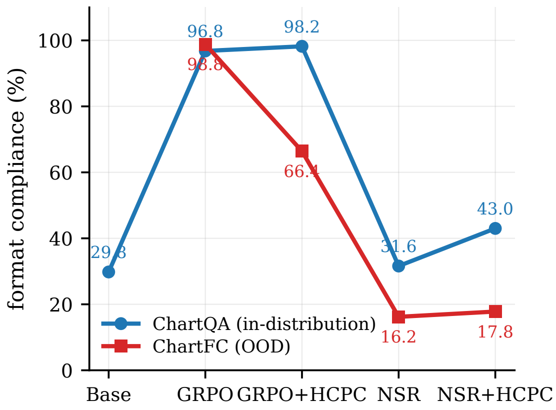

Figure 4: Overall format compliance per method on ChartQA (in-distribution) and ChartFC (OOD). NSR collapses on both; GRPO+HCPC holds ID but degrades OOD; GRPO is the only configuration that keeps format near ceiling under distribution shift.

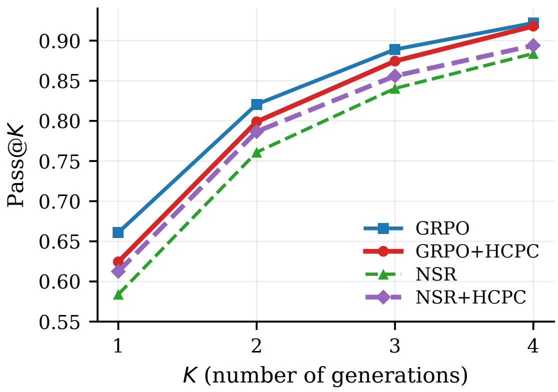

Figure 5: Pass@K on ChartFC (OOD, n=500). Plain NSR (green) sits below all other configurations at every K; adding HCPC (purple) closes most of the gap to GRPO. BASE was not evaluated on ChartFC.

table extraction stops actively hurting; the format gain is marginal. Plain NSR is the weakest OOD configuration on every metric (Table 1), and HCPC closes most of that gap (Figure 5). How the gap is closed turns out to matter more than the gap itself.

Format/accuracy dissociation. The way HCPC mitigates NSR's damage is unusual: NSR+HCPC reaches Pass@1 = 0.632, statistically tied with GRPO (0.638; paired-bootstrap CI $[-0.046, +0.060]$, p=0.85), while format compliance remains collapsed at 17.8% (vs. GRPO's 98.8%). HCPC under NSR recovers GRPO-level OOD accuracy without recovering format: the cleanest evidence in the paper that format compliance and answer accuracy are separable channels in chart RLVR.

Why HCPC behaves so differently under GRPO and NSR. Under GRPO, HCPC acts as designed: the bonus raises correct-path rollouts' rewards, lifts their advantage $(R_i - \bar{R})/s_R$, and reinforces the correct path. This standard positive-gradient interpretation accounts for the ChartQA results (§5.2).

Under NSR, the same bonus operates through a different channel. HCPC's bonus lands on exactly the correct-path rollouts that NSR masks, so it never reaches the policy as a direct positive update. But TRL computes the GRPO advantage before the NSR mask is applied, using the group mean $\bar{R}$. Adding R_{HCPC} to correct rollouts raises $\bar{R}$, making the surviving wrong rollouts' advantages more negative. The bonus thus reshapes the baseline that wrong rollouts are penalized against, pushing the policy harder away from wrong-path behavior. The mechanism is *baseline shift, not direct reinforcement*: HCPC under NSR strengthens the negative gradient on wrong rollouts without giving any positive gradient to correct ones. This matches what we observe: answer accuracy OOD recovers (wrong-rollout suppression strengthens) while structured-output emission, which needs positive gradient on correct rollouts, stays collapsed.

On ChartQA, adding HCPC on top of NSR also partially restores format (31.6% → 43.0% overall, with the recovery concentrated in ⟨table⟩: 72.2% → 83.2%, Table 2). In-distribution, format partially recovers. Out-of-distribution, accuracy recovers fully while format does not. This asymmetry makes the channel-separability claim concrete.

6 Conclusion

We introduced HCPC, a cross-rollout bonus rewarding consistent extraction and diverse reasoning on the correct-path subset of a group. On GRPO it yields more robust correct groups without sacrificing Pass@1. Porting NSR, from text-only math RLVR, into chart RLVR collapses format compliance: structural rewards saturate fast enough that well-formed rollouts cross NSR's threshold regardless of answer quality. HCPC on top of NSR recovers GRPO-level OOD accuracy at near-zero format, direct evidence that format and answer accuracy are separable channels. Reward-rule asymmetries from single-reward domains can interact destructively with added reward components, making multi-component reward design a first-class concern when transferring RLVR across modalities.

Limitations

We use a single 3B-parameter backbone (Qwen2.5-VL-3B), a 1,000-sample training subset, $K{=}4$ rollouts, and a 500-sample test slice per benchmark. Compute constraints prevented us from training on the full Chart-RVR CoT mixture (34,194 samples), evaluating on the full ChartQA and ChartFC test sets, or running EvoChart (?) and ChartQAPro (?) as style-OOD benchmarks distinct from the joint task-and-style probe ChartFC provides. Scaling to the full training data, full test sets, and a style-only OOD benchmark is our highest-priority follow-up.

A Reproducibility

Training and evaluation code is included with the submission. Hyperparameters and the exact reward configuration used for every run are specified in the training configs (`configs/experiment.py`); the expected output schema is in §B. Pass@K uses the unbiased estimator of ?. Cross-rollout similarities use MiniLM-L6 embeddings. Paired-bootstrap CIs use $n_{\text{boot}}{=}10{,}000$ resamples; McNemar tests use the two-sided exact binomial form on the discordant pair.

B Prompt Template and Expected Output Format

The model is prompted to emit responses in a structured XML template. The prompt instructs the model to wrap its reasoning in `<think>...</think>`, identify the chart type, extract the underlying data table, perform step-by-step reasoning, and finally emit the answer in `<answer>...</answer>`. The expected output schema is:

```
<think>
<type>...</type>
<table>{"columns": [...], "rows":
↪   [...]}</table>
<step-1>...</step-1>
<step-2>...</step-2>
...
</think>
<answer>...</answer>
```

The per-tag emission rate metric (§4.2) and the format reward R_{fmt} are both defined relative to this schema. `<think>` and `<table>` are not natively emitted by Qwen2.5-VL-3B-Instruct without training; the format reward provides the gradient that teaches them.

C Training Dataset Example

Each training instance is a tuple of (image, query, ground-truth label, chart type, reference table, expected reasoning trace). The prompt sent to the model is the system message from §B together with the chart image and the query. The reward signals defined in §3.1 are computed against the ground-truth label, chart type, and reference table; the reference reasoning trace is used only for the (disabled) process-conformity reward, not in our reported runs. Figure 6 shows the image; the remaining fields for this instance are below.

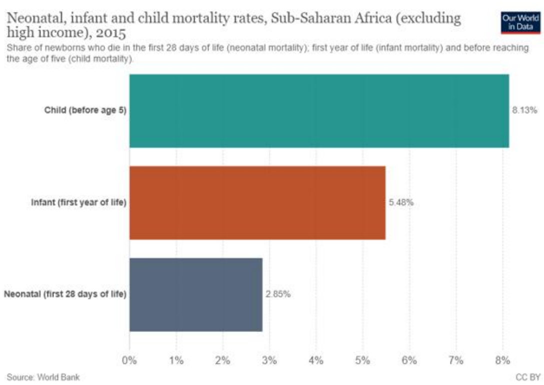

Figure 6: Chart image for the dataset example below. Source: Our World in Data / World Bank, reproduced under CC-BY as it appears in the training mixture.

Query.

```
What is the value of Child bar?
```

Ground-truth label.

```
8.13
```

Chart type.

```
bar
```

Reference table.

```
{
  "columns": ["Category", "Value"],
  "rows": [
    ["Child (before age 5)",
    ↪   "8.13%"],
    ["Infant (first year of life)",
    ↪   "5.48%"],
    ["Neonatal (first 28 days of
    ↪   life)","2.85%"]
  ]
}
```

Reference reasoning trace.

```
<step-1>: Identify the relevant bar for 'Child' in the chart.
<step-2>: Read the value associated with the 'Child' bar.
<step-3>: Verify the value matches the percentage label.
<step-4>: Confirm the value is 8.13%.
<step-5>: Verify the final answer for hallucinations.
```

Dataset schema. The training mixture is a Hugging Face dataset with the following columns: `image` (chart PNG), `query` (natural-language question), `label` (ground-truth answer string), `prompt` (assembled system prompt + query), `reasoning` (reference reasoning trace), and `chart_type` (categorical label). Per-rollout outputs are parsed against the expected schema in §B to extract the predicted answer, table, and chart type for reward computation.

D Per-Tag Format Compliance, Both Benchmarks

Table 2 in the main body reports per-tag emission rates for ChartQA. Table 3 extends this to both benchmarks. BASE was not evaluated on ChartFC.

Method	think	answer	type	table	order
ChartQA (in-distribution)					
BASE	52.2	91.2	96.6	77.2	81.6
GRPO	97.4	97.6	100	98.6	97.4
GRPO+HCPC	98.8	98.2	100	99.6	98.6
NSR	57.4	94.8	96.6	72.2	85.4
NSR+HCPC	57.8	91.6	94.6	83.2	84.0
ChartFC (out-of-distribution)					
GRPO	99.0	99.0	100	99.6	99.0
GRPO+HCPC	78.0	92.4	96.6	87.2	79.0
NSR	51.4	91.0	96.2	55.6	56.0
NSR+HCPC	49.2	90.2	95.6	61.4	56.0

Table 3: Per-tag emission rate (%) on both benchmarks. On ChartFC, the <think> and <table> tags collapse hardest under NSR variants (51.4–61.4%, vs. GRPO's 99.0–99.6%). GRPO+HCPC also degrades OOD but milder ($\sim$78–87%), in line with the OOD analysis in §5.2.

E Pairwise Statistical Comparisons

We computed paired-bootstrap 95% confidence intervals (n_{boot}=10,000) and exact two-sided McNemar p-values for all 20 method-pair $\times$ Pass@K $\times$ benchmark comparisons. Among the trained configurations (GRPO, GRPO+HCPC, NSR, NSR+HCPC), two pairwise gaps reach $p < 0.05$: GRPO vs. NSR on ChartFC Pass@4 (+3.8 pp, p=0.034; reported in §5.3) and GRPO vs. NSR+HCPC on ChartQA Pass@1 (+5.6 pp, p=0.029). All comparisons involving the untuned BASE model are significant on Pass@1 ($p < 0.05$). All other pairs, including the headline GRPO vs. GRPO+HCPC comparison (-0.8 pp on Pass@1, p=0.80; $+1.2$ pp on Pass@4, p=0.52), do not reach significance at α=0.05.

F HCPC Firing-Rate Distribution

HCPC's group bonus is zero when fewer than two rollouts pass the correct-path filter ($|G^+| < 2$). The empirical distribution of $|G^+|$ across the 500 ChartQA test groups for the trained GRPO+HCPC model is:

$\lvert G^+\rvert$	0	1	2	3	4
fraction	17.2%	11.6%	14.6%	15.8%	40.8%

HCPC fires (non-zero bonus) on groups with $|G^+| \geq 2$, i.e. 71.2% of groups; the remaining 28.8% are silent. Most firing groups are concentrated at the high end of the distribution ($|G^+|=4$ alone accounts for 40.8%), so the bonus is mostly applied where the policy already produces multiple correct rollouts and benefits from cross-rollout consistency and diversity signals. The 17.2% all-wrong rate reflects strong positive correlation of correctness within a group, as expected when all K=4 rollouts share the same chart image and question.

G Δ_{tbl} on the Common-Table Subset

To rule out the confound that Δ_{tbl} is computed on different denominators per method, we recompute the table-conditioned arm on the intersection of samples where both methods produce at least one parseable-table rollout. On the 498 ChartQA samples where both GRPO and GRPO+HCPC have a parseable table available, $P(\hat{y} = y^* \mid \text{tbl})$ is 0.623 for GRPO and 0.634 for HCPC: indistinguishable. The two methods reach the same answer accuracy when a parseable table is produced. The Δ_{tbl} gap in Table 1 therefore arises entirely from the no-table arm. On the full slice, $P(\hat{y} = y^* \mid \text{no tbl})$ is 0.360 for GRPO and 0.490 for HCPC, so HCPC is more robust to extraction failures rather than more dependent on the extracted table for correctness.

H Per-Answer-Type Breakdown

We split ChartQA test samples by ground-truth answer type and compare GRPO against GRPO+HCPC on Pass@1. Differences are within exact-McNemar noise on numeric (n=302, -1.7 pp, p=0.63), boolean (n=75, $+4.0$ pp, p=0.71), and string (n=114, -6.1 pp, p=0.30). HCPC does not provide a category-specific accuracy advantage at this scale; the rollout-set robustness gains reported in §5.2 are uniform across answer types rather than concentrated on any single category.

I Qualitative Example: Format Collapse in Action

Illustration of the NSR collapse and HCPC partial recovery on a single ChartQA sample. The question is *"How many categories are there in the chart?"* with ground truth $y^* = 4$. We show rollout 0 from three methods.

GRPO (correct, well-formed).

```
<think><type>pie</type><table>{..., "rows":
↪  [["Asians","63"],...]}</table>
<step-1> The chart is a pie chart...
<step-4> Based on the information provided,
↪  there are four distinct categories...
</think>
<answer>4</answer>
```

NSR (format-collapsed; arrives at right reasoning but wrong answer surface form).

```
<think>First, I'll identify the type of chart
↪  and the underlying data table...
<step-1> The chart shows four categories:
↪  Asians, Whites, Blacks, and Hispanics. ...
Final answer:
Four
</think>
<answer>Four</answer>
```

NSR's rollout contains no `<table>` tag, no `<type>` tag, and emits the reasoning steps inline rather than as structured items. The answer is also expressed in word form ("Four") rather than the numeric form the eval expects ("4"), so the strict-match metric marks it incorrect despite the reasoning being substantively right.

NSR+HCPC (table tag recovered; answer still in word form).

```
<think><type>bar</type><table>{"columns":
↪  ["Country","Bar Color"], "rows":
↪  [["Asians","..."],...]}</table>
<step-1>From the chart...
<step-3>... there are four distinct
↪  categories.
</think>
<answer>Four</answer>
```

HCPC's group-level bonus on C_{table} pulls NSR's policy back toward emitting a `<table>` tag (table fraction $72.2\% \rightarrow 83.2\%$ on ChartQA, Table 2). The chart type is incorrectly identified (`bar` instead of `pie`), illustrating that HCPC's partial format recovery is not the same as a full quality recovery, and the answer is still "Four" in word form. This is exactly the dissociation pattern of §5.4: format is partly restored, but the answer-surface alignment that drives strict-match accuracy is governed by a separate channel.